Drive-law noise reduction: fw v35 vs fw v37

Scale-car DC balancer board, Pi rosbag comparison. Ricky Tan, 2026-11-06.

1. Summary

The fw v37 drive law lowers the drive-loop crossover from 2.76 Hz to 1.04 Hz. On the vehicle, it reduces the once-per-revolution (1x) flywheel line to 0.25–0.28 of its fw v35 level on the bus currents, the bus voltage and the FC share ratio. The design ratio is 0.29. Content above about 7 Hz is unchanged.

2. Data and method

Two runs of the same SDP drive cycle are compared: `rosbag_sdp_2026-10-30_01-49-38` (fw v35) and `rosbag_sdp_2026-11-05_20-09-52` (fw v37 rev 2). The share controller and all other run-path code are identical in the two builds. Statistics use only samples with both sources on the bus and $I_{tot} > 0.3$ A. The analysis excludes ± 0.5 s around each H-20 purge and $-0.2/+1.0$ s around each share-command step. Duplicate telemetry rows are removed, which gives a sample rate of 47.62 Hz. The share is $s = I_{fc}/I_{tot}$.

3. Results

Table 1 gives the 1x amplitude from a tracked-phase lock-in at 1.8–2.9 m/s.

1x amplitude (peak)	fw v35	fw v37	Ratio
I_{tot}	213 mA	54 mA	0.25
I_{batt}	187 mA	47 mA	0.25
I_{fc}	34 mA	9 mA	0.27
V_{bus}	14.2 mV	3.9 mV	0.28
Share ratio	0.032	0.008	0.25
v_{actual}	18.6 mm/s	12.1 mm/s	0.65

Table 1. 1x line at cruise (1160 / 1264 samples).

The bus-side 1x gain I_{tot}/v_{actual} fell to 0.33, against the design controller-gain ratio of 0.34, so the board runs the new law as designed. The v_{actual} 1x fell more than the new law alone predicts (0.65 against 0.85). The once-per-revolution disturbance was therefore smaller in the fw v37 run, for a reason the bags do not show.

The bus voltage follows $V_{bus} \approx V_0 - Z\, I_{tot}$, where Z is the bus impedance that the two droops set in parallel. Table 2 gives Z from the $I_{tot} \rightarrow V_{bus}$ cross-spectrum and the fraction of the V_{bus} variance that I_{tot} explains.

Run	Band	$\lvert Z \rvert$ (mΩ)	V_{bus} rms (mV)	Explained by I_{tot}
fw v35	2–6 Hz	57	10.8	67 %
fw v35	6–24 Hz	41–47	7.0–8.7	9–12 %
fw v37	2–6 Hz	41	5.4	21 %
fw v37	6–24 Hz	–	5.5–8.1	5–7 %
Battery only	2–6 Hz	159	23.1	92 %

Table 2. Bus impedance and the droop-driven part of V_{bus}, clean cruise windows.

At the 1x line, V_{bus}/I_{tot} is 67 mΩ before and 73 mΩ after. The V_{bus} 1x reduction is therefore the I_{tot} 1x reduction passed through an unchanged droop. This impedance is about 0.2 of the design droop of 0.30–0.36 Ω. In fw v37, most of the remaining V_{bus} noise is not coherent with I_{tot}, so it does not come through the droop. Its origin is not yet determined.

Figure 1 shows the cruise spectra. The 1x peak falls by about one decade in power on every channel, including the share ratio. Above about 7 Hz the two runs overlap.

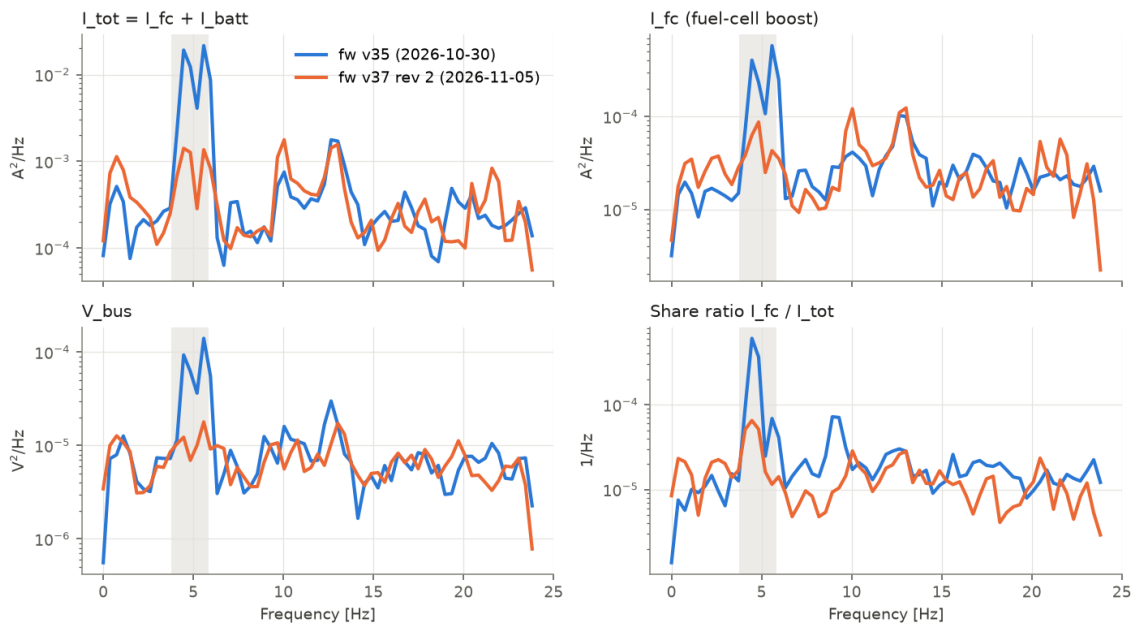


Figure 1. Cruise PSDs (v $>$ 1.8 m/s), fw v35 (blue) and fw v37 (orange). Grey band: 1x frequency at 1.8–2.8 m/s.

The share ratio appeared not to improve because two terms that the drive law does not touch dominate it: the H-20 purges and the SDP command steps. Figure 2 shows the share 2–6 Hz rms as each term is removed. With both removed, the share noise falls from 0.021 to 0.010.

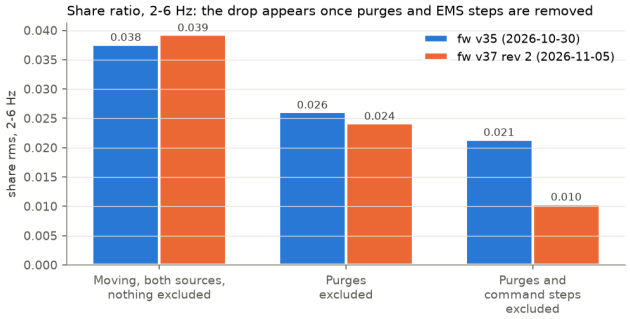


Figure 2. Share-ratio rms, 2–6 Hz, under successive masks.

4. Conclusions

- The lower crossover reduced the 1x line to 0.25–0.28 on all bus channels and the share ratio, in agreement with the design.
- V_{bus} at 1x is the droop image of I_{tot} through about 70 mΩ. Most of the remaining V_{bus} noise is not droop-driven.
- The share-ratio reduction is real. Purges and command steps hide it in unmasked data.
- Content above 7 Hz did not change. Its largest share component is a flat floor of about 0.012 rms, of undetermined origin. A 1 kHz SD-card log with synchronous current reads will determine whether it is a sampling artefact.
- The tracking cost is visible: the speed-error rms below 0.7 Hz increased from 5.0 to 9.8 mm/s.